📘 Lecture 4 – Testing

30 Exam-Style MCQs (Dr. Ehab Essa Style)

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1. Testing is part of which broader process?

a) Debugging
b) Validation
c) Compilation
d) Deployment

Answer: b

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2. The main goal of validation is to:

a) Remove code
b) Increase confidence
c) Improve speed
d) Reduce memory

Answer: b

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3. Which validation technique involves formal proof?

a) Testing
b) Code review
c) Verification
d) Debugging

Answer: c

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4. Running a program with selected inputs is called:

a) Verification
b) Inspection

c) Testing
d) Analysis

Answer: c

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5. Exhaustive testing is:

a) Efficient
b) Required
c) Infeasible
d) Automatic

Answer: c

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6. Why is exhaustive testing impossible?

a) High cost
b) Infinite inputs
c) Large input space
d) Slow execution

Answer: c

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7. Random testing is weak because software defects are:

a) Uniform
b) Continuous
c) Discrete
d) Predictable

Answer: c

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8. “Just try it and see” testing is called:

a) Exhaustive
b) Random
c) Haphazard
d) Boundary

Answer: c

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9. In test-first programming, what is written first?

a) Code
b) Tests
c) Debugger
d) Driver

Answer: b

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10. Test-first programming starts with writing:

a) Code
b) Tests
c) Specification
d) Comments

Answer: c

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11. The specification mainly describes:

a) Implementation
b) Output only
c) Input-output behavior
d) Code structure

Answer: c

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12. Writing tests early helps detect:

a) Compiler bugs
b) Spec problems
c) Hardware errors
d) IDE issues

Answer: b

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13. Dividing input space into subdomains is called:

a) Coverage
b) Partitioning
c) Debugging
d) Profiling

Answer: b

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14. Partitioning aims to achieve:

a) Maximum tests
b) Random coverage
c) Maximum coverage
d) Full execution

Answer: c

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15. Bugs frequently occur at:

a) Random inputs
b) Middle values
c) Boundaries
d) Typical cases

Answer: c

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16. Which is a common boundary?

a) Random number
b) Empty input
c) Normal value
d) Average case

Answer: b

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17. An off-by-one error is related to:

a) Compilation
b) Boundaries
c) Syntax
d) Typing

Answer: b

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18. Blackbox testing relies on:

a) Implementation
b) Source code
c) Specification
d) Debugger

Answer: c

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19. Whitebox testing uses knowledge of:

a) Requirements
b) Specification
c) Implementation
d) User behavior

Answer: c

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20. Choosing tests based on loops and branches is:

a) Blackbox
b) Random
c) Whitebox
d) Exhaustive

Answer: c

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21. Statement coverage measures:

a) Paths
b) Branches
c) Executed statements
d) Input values

Answer: c

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22. Branch coverage requires:

a) All statements run
b) All paths run
c) True and false branches
d) Random inputs

Answer: c

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23. Path coverage is usually:

a) Easy
b) Required
c) Infeasible
d) Automatic

Answer: c

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24. A unit test focuses on:

a) Whole program
b) Multiple modules
c) Single module
d) User behavior

Answer: c

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25. Integration testing checks:

a) Syntax
b) Single method

c) Module interaction
d) Code style

Answer: c

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26. Automated testing means tests are:

a) Manual
b) Random
c) Automatically checked
d) Informal

Answer: c

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27. Re-running tests after changes is called:

a) Unit testing
b) Regression testing
c) Integration testing
d) Debugging

Answer: b

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28. A test added after fixing a bug is a:

a) Unit test
b) Boundary test
c) Regression test
d) Random test

Answer: c

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29. JUnit is mainly used for:

a) Debugging
b) Profiling
c) Automated testing
d) Compilation

Answer: c

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30. Test-first debugging starts by:

a) Fixing code
b) Writing test
c) Reviewing code
d) Running program

Answer: b

📘 Lecture 4 – Testing

Essay Questions (Dr. Ehab Essa Style)

📝 Essay Question 1

Define software testing. Explain its role in validation.

Model Answer:

Software testing is the process of **running a program with selected test cases** to find defects. Testing is a key part of **validation**, whose goal is to **increase confidence** that the software behaves as intended.

Validation includes:

- **Verification** (formal reasoning)
- **Code review** (human inspection)
- **Testing** (execution-based checking)

Testing does not prove correctness but increases confidence in software quality.

📝 Essay Question 2

Explain input space partitioning and boundary testing. Why are they important?

Model Answer:

Input space partitioning divides the input domain into **subdomains** where the program behaves similarly.
One representative test is selected from each subdomain to achieve **maximum coverage** with fewer tests.

Boundary testing focuses on **edge cases** such as zero, empty inputs, and maximum values.
Boundaries are important because many bugs, such as **off-by-one errors**, occur at boundaries.

Together, they help design **effective test cases**.

📝 Essay Question 3

Compare blackbox testing and whitebox testing.

Model Answer:

Blackbox testing selects test cases based on the **specification**, without considering implementation details.
It focuses on **input-output behavior**.

Whitebox testing selects test cases based on the **implementation**, including branches, loops, and paths.
It focuses on **code structure**.

Blackbox testing checks correctness against requirements, while whitebox testing improves **code coverage**.